

Reflection: Mini Calculator Application

Design Decisions and Rationale : For the Mini Calculator exercise, the primary design decision was to strictly separate the graphical user interface from the underlying mathematical operations. This was achieved by creating two distinct classes: CalculatorGUI and CalculatorLogic. This separation of concerns ensures that the application adheres to fundamental object-oriented programming principles. By decoupling the calculation engine from the Swing components, the codebase becomes more modular, easier to maintain, and highly extensible. The user interface was built utilizing Java Swing , specifically employing BorderLayout for the overall frame and GridLayout to systematically arrange the calculator buttons, ensuring a clean, intuitive, and user-friendly layout.

Main Functionalities : The application successfully implements all required calculator functionalities. It features a non-editable display screen to clearly show current inputs and results , along with a complete set of number buttons from 0 to 9. Users can perform standard basic arithmetic operations, including addition, subtraction, multiplication, division, and modulo. Furthermore, control buttons such as "C" (Clear) and "<-" (Delete) are fully functional, allowing users to reset the calculator entirely or remove the most recently entered character. The application is also designed to robustly handle invalid inputs, safely catching exceptions like division by zero or incomplete expressions to prevent crashes.

Brief Explanation of the Implementation : The implementation heavily relies on event-driven programming. Within the CalculatorGUI class, an ActionListener is attached to every button. When a user interacts with a button, the input is processed to determine if it is a digit, an operator, or a control command. Upon pressing the "=" button, the GUI extracts the operands and passes them to the CalculatorLogic class. The logic class utilizes a switch expression to compute the result and returns it. Any ArithmeticException or NumberFormatException thrown during calculation is caught gracefully, displaying an "Error" message on the screen and ensuring the application runs smoothly without errors.